SpaceX Launch Success Analysis

Capstone Project - Data Science

GitHub Repository:

https://github.com/youssefeltabee/spacex-capstone

All notebooks and Python files available in repository

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Executive Summary

Analyzed 90 SpaceX launches from historical dataset

Overall success rate: 66.67% (60 successes, 30 failures)

Key findings:

- KSC LC-39A (77.27%) and VAFB SLC-4E (76.92%) outperform CCAFS SLC-40 (60.0%)
- GTO missions show lowest success rate (51.85%)
- VLEO/SSO missions highly successful (85.71% / 100%)
- Predictive model (Random Forest) achieved AUC = 0.926
- Top predictive features: ReusedCount, Legs, GridFins

Introduction & Problem Statement

Problem Statement:

Analyze historical SpaceX Falcon 9 launch data to identify factors associated with launch success and provide actionable insights.

Research Questions:

1. What is the dataset-level success rate?
2. Which launch sites have higher/lower success rates?
3. Are booster categories associated with different success rates?
4. Does payload mass affect success?
5. Can we predict launch success using machine learning?

Data Collection & Wrangling Methodology

Data Sources (Capstone folder):

- dataset_part_2.csv - Main cleaned dataset (90 launches)
- spacex_launch_dash.csv - Booster version categories
- dataset_part_1.csv, dataset_part_3.csv - Additional features
- Spacex.csv - Raw launch data
- my_data1.db - SQLite database

Wrangling Steps:

- Harmonized column names and data types
- Mapped FlightNumber to Booster Version Category
- Handled missing values (PayloadMass, dates)
- Created binary Class variable (1=Success, 0=Failure)
- Generated dummy variables for categorical features

EDA & Interactive Visual Analytics Methodology

Exploratory Data Analysis:

- Summary statistics and distributions
- Success rate by launch site, orbit, booster category
- Payload mass analysis (boxplots, means)
- Time-series analysis (rolling success rates)

Interactive Visual Analytics:

- Folium interactive map with launch site markers
- Plotly Dash dashboard (spacex-dash-app.py)
- Dynamic filtering by launch site, orbit, booster type
- Hover tooltips with launch details

Predictive Analysis Methodology

Machine Learning Approach:

- Features: PayloadMass, Flights, GridFins, Reused, Legs, Block, ReusedCount
- Categorical dummies: Orbit_*, LaunchSite_*, Serial_*
- Train/test split: 70/30 with stratification
- Models evaluated:

- Logistic Regression (baseline)
- Random Forest Classifier

- Metrics: AUC-ROC, precision, recall, F1-score
- 10-fold cross-validation for robust evaluation

EDA Results - Visualizations

Figures (see generated PNG files):

- Figure 1: Success Rate by Launch Site (horizontal bar)
- Figure 2: Success Rate by Orbit (top groups)
- Figure 3: Payload Mass by Outcome (boxplot)
- Figure 4: Rolling Success Rate over Time

Key insights from visualizations:

- Launch site performance varies significantly
- Orbit type strongly correlates with success
- Payload mass distribution differs by outcome

EDA Results - SQL Queries

SQL Analysis (my_data1.db):

1. Success Rate by Launch Site:
   - CCAFS SLC 40: 60.0% (55 launches)
   - KSC LC 39A: 77.27% (22 launches)
   - VAFB SLC 4E: 76.92% (13 launches)

2. Success Rate by Orbit:
   - GTO: 51.85% (27 launches)
   - ISS: 61.90% (21 launches)
   - VLEO: 85.71% (14 launches)

3. Payload Analysis:
   - Success mean: 6,765 kg
   - Failure mean: 4,785 kg

Queries saved in: spacex_queries.sql

Interactive Map Results - Folium

Interactive Launch Site Map:

- Created using Folium with MarkerCluster
- Green markers = Successful launches
- Red markers = Failed launches
- Click markers for launch details
- Map saved as: spacex_launch_map.html

Launch Site Coordinates:

- CCAFS SLC 40: 28.56°N, 80.58°W
- KSC LC 39A: 28.61°N, 80.60°W
- VAFB SLC 4E: 34.63°N, 120.61°W

Plotly Dash Dashboard Results

Interactive Dashboard Features:

- File: spacex-dash-app.py
- Visualizes launch data from spacex_launch_dash.csv
- Interactive components:

- Payload mass histogram with range slider
- Launch site success pie chart
- Booster version category analysis
- Time-series of launch success

Dashboard allows stakeholders to:

- Filter launches by site, orbit, booster type
- View real-time success metrics
- Export filtered data for further analysis

Predictive Analysis Results

Model Performance:

- Random Forest AUC: 0.926
- Logistic Regression AUC: 0.815
- Top Features: ReusedCount (0.24), Legs (0.18), GridFins (0.16)
- Random Forest Precision (Success): 0.88
- Random Forest Recall (Success): 0.83

Model Figures Generated:

- feature_importance.png - Top 15 features
- roc_curves.png - ROC curves for both models
- confusion_matrix.png - Confusion matrix (Random Forest)

Conclusion

Key Findings:

1. Launch site and orbit type significantly impact success rates
2. GTO missions historically riskier (51.85% success)
3. Modern booster versions (FT, B4, B5) show improved performance
4. Machine learning models can predict success with AUC > 0.92

Recommendations:

- Focus risk mitigation on GTO missions
- Use model predictions for pre-launch risk assessment
- Continue collecting data for improved model accuracy
- Develop real-time dashboard for mission planning

Creativity & Innovative Insights

Beyond the Template - Added Value:

1. Interactive Folium Map
   - Cluster markers for dense launch regions
   - Popup details with payload and outcome

2. Comprehensive Predictive Modeling
   - Feature importance analysis with visualization
   - ROC curves and confusion matrices
   - Model comparison (Logistic vs Random Forest)

3. Full Git Repository
   - All scripts, notebooks, and outputs version-controlled
   - Reproducible analysis pipeline

4. Multi-format Delivery
   - PDF report, PowerPoint, Jupyter notebook
   - Interactive HTML map and SQLite database

GitHub Repository & Resources

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https://github.com/youssefeltabee/spacex-capstone

Contents:

- Jupyter Notebooks:

  - Capstone_findings.ipynb (EDA notebook)
  - edadataviz.ipynb (EDA with visualizations)
  - SpaceX_Machine Learning Prediction_Part_5.ipynb

- Python Scripts:

  - generate_capstone_reports.py
  - predictive_model.py
  - folium_map.py
  - spacex-dash-app.py

- Data Files: dataset_part_1/2/3.csv, Spacex.csv, etc.
- Outputs: PDF report, PPTX, HTML map, model figures